The torque of the force $\vec{F}$ about O is, $\vec{\tau} = \vec{r} \times \vec{F}$. The component of the torque $\vec{\tau}$ along the axis is the torque about the axis. To find it, we should first find the vector $\vec{\tau} = \vec{r} \times \vec{F}$ and then find the angle ϕ between $\vec{\tau}$ and the axis AB. (Remember here, the force $\vec{F}$ is not on the plane ABP). The torque about the axis AB is the parallel component of the torque along the axis AB, which is $|\vec{r} \times \vec{F}|$ cos ϕ. The torque perpendicular to the axis AB is $|\vec{r} \times \vec{F}|$ sin ϕ.

The torque about the axis will rotate the object about the axis and the torque perpendicular to the axis will turn or tilt the axis of rotation itself. When both components exist simultaneously on a rigid body, the body will have a *precession*. One can witness the precessional motion in a spinning top when it is about to come to rest as shown in Figure 5.9.

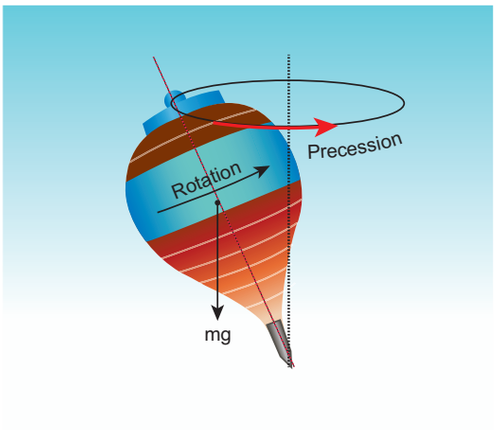


Figure 5.9 Precession of a spinning top

Study of precession is beyond the scope of the present course of study. Hence, it is assumed that there are constraints to cancel the effect of the perpendicular components of the torques, so that the fixed position of the axis is maintained. Therefore, perpendicular components of the torque need not be taken into account.

For the rest of the lesson, we consider rotation about only fixed axis. For this we shall,

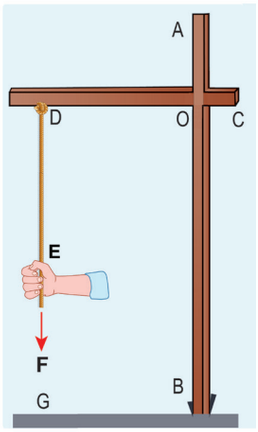
1. *Consider forces that lie only on planes perpendicular to the axis (without intersecting in the axis).*
2. *Consider position vectors that are only perpendicular to the axis.*

Note

- Forces parallel to the axis will give torques perpendicular to the axis of rotation and need not be taken into account.
- Forces that intersect (pass through) the axis cannot produce torque as r = 0.
- Position vectors along the axis will result in torques perpendicular to the axis and need not be taken into account.

EXAMPLE 5.10

Two mutually perpendicular beams AB, CD, are joined at O to form a structure which is fixed to the ground firmly as shown in the Figure. A string is tied to the point D and its free end E is pulled with a force $\vec{F}$. Find the magnitude and direction of the torque produced by the force,



(i) about the points E, D, O and B,

(ii) about the axes DE, CD, AB and BG.

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